

Final Project Math 2406H Spring 2026

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May 11, 2026

1 Project Overview

A PID controller is the most widely used control function in many applications. Its desirability comes from its ability to adapt to practically any system and its relative ease of implementation. It functions by adjusting the control input into a system based on the system's error. This input is calculated using three components: the term directly proportional to the error, the term proportional to the integral of the error, and the term proportional to the derivative of the error, PID for short. The basic equation that we will use for this project is as follows

$$u(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt} \quad (1)$$

where $u(t)$ is the calculated control signal given to the system, t is the time, and $e(t)$ is the error of the system at the given time. The term K_p represents the constant for the part directly proportional to the error $e(t)$, the term K_i represents the constant for the part proportional to the integral of the error over time, and the term K_d represents the constant for the part proportional to the derivative of the error.

The goal of this project is to model a simple PID controller for a classic example, the inverted pendulum. First, we will derive the position equations using Lagrangian mechanics, this will give us the system of non linear equations that describe the pendulums movement. Then, to determine whether control is really necessary for this system, we will convert this non-linear system in to a first order system of linear equations and perform a stability analysis around the upright position. Finally, we will select a cost function and use the method of gradient descent to optimize the values for K_p , K_i , and K_d . This is an especially interesting goal because picking the optimal values for these constants is often done manually (Qiao 2026; *Lecture 18: Systems and Controls* 2026; *Inverted Pendulum* 2026a; *Inverted Pendulum* 2026b; *Inverted Pendulum* 2026c; Murari, Ahmad, and Kumar 2019; Saharoy 2026; *Inverted Pendulum: System Modeling* 2026).

2 Modeling the System

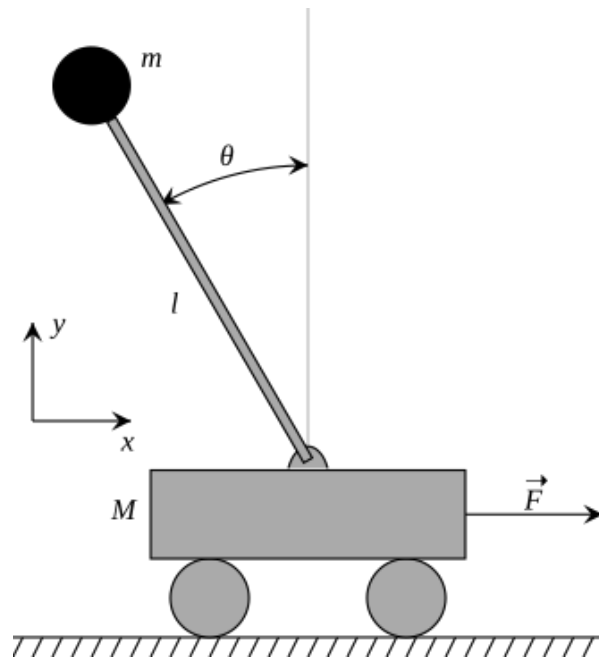


Figure 1: "Schematic drawing of an inverted pendulum on a cart." by Krishnavedala is licensed under CC0 1.0

The inverted pendulum system is shown above. The position of the pendulum is determined by that of the cart; to change the position of the pendulum, one must apply a force to the cart. The PID controller is necessary to calculate the correct force to apply to the cart to keep the pendulum balanced.

As with any thing in physics we will make somewhat hefty assumptions to be able to model the system easily. The assumptions we will make for our system are the following.

1. The pendulum can be treated as a massless rod, and the bob of the pendulum as a point mass. This allows the moment of inertia for the pendulum to simply be ml^2 and for the movement of the rod itself to be ignored. This assumption will cause the system to be more responsive than if the rod of the system was not massless.
2. Friction can be entirely ignored. This allows the pendulum to rotate freely about the axis and the cart to move with no resistance. This will cause the system to be less responsive to external forces.
3. Similarly, air resistance can be ignored. This will likely have little effect on the system as it will not be moving that fast.
4. The cart can move infinitely along its track. In reality the x position of the cart would have to be bounded by some constraints which would require dedicated logic for these edge cases. To simplify, the cart is allowed to move as it pleases. This does not effect the dynamics of the system or the controller as it is the same if there were bounds and the cart was simply far away from these bounds.
5. The calculated force will be instantaneously applied to the system. In reality, it would take time to produce the necessary force. This will cause the system to be much more easily controlled.

With these assumptions, the system is much more reliably and easily simulated. Now we will derive the equations of motion to use for the model. As a note $\dot{x} = \frac{dx}{dt}$, for convention and readability. To solve for the equations we need, it is easiest to use the Lagrangian, defined as $\mathcal{L} = E_k - E_p$.

To start, total kinetic energy of the cart + pendulum system is the sum of the kinetic energy of the cart and the pendulum, which is

$$E_k = \frac{1}{2}(M + m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{\theta}\dot{x}\cos\theta \quad (2)$$

Next, the total potential energy is simply that of the pendulum as the cart does not change height, giving

$$E_p = mgl\cos\theta \quad (3)$$

Combining equations 2 and 3 with the Lagrangian gives

$$\mathcal{L} = \frac{1}{2}(M + m)\dot{x}^2 + \frac{1}{2}ml^2\dot{\theta}^2 + ml\dot{\theta}\dot{x}\cos\theta - mgl\cos\theta \quad (4)$$

Now we obtain the actual equations for the force on the cart and we apply the Euler-Lagrange equation, defined as $\frac{d}{dt}\frac{\partial\mathcal{L}}{\partial\dot{q}_i} - \frac{\partial\mathcal{L}}{\partial q_i} = Q_i$ where Q_i denotes the actual force on the system for that coordinate with $Q_x = F$, the force on the cart, and $Q_\theta = 0$, because no external torque acts on the actual hinge of the pendulum.

Solving the Euler-Lagrange equation for $q = x$ gives

$$(M + m)\ddot{x} - ml\ddot{\theta}\cos\theta + ml\dot{\theta}^2\sin\theta = F(t) \quad (5)$$

and solving for $q = \theta$ gives

$$l\ddot{\theta} - \ddot{x}\cos\theta - g\sin\theta = 0 \quad (6)$$

2.1 linearization

The goal of linearization is to be able to definitively say the system is unstable at its upright equilibrial $\theta = 0$. To be able to perform a stability analysis of the point of interest the we must convert these two equations into a first order system and then linearize it around $\theta = 0$.

To perform this linearization, we define four variables as such

$$\begin{aligned} x_1 &= x \\ x_2 &= \dot{x} \\ x_3 &= \theta \\ x_4 &= \dot{\theta} \end{aligned}$$

Differentiating this gives the following first order system $\dot{\mathbf{x}} = f(x, F)$ where $\mathbf{x}$ is defined as such

$$\begin{aligned} \dot{x}_1 &= \dot{x} \\ \dot{x}_2 &= \ddot{x} = \frac{F + mg\sin(x_3)\cos(x_3) - mlx_4^2\sin(x_3)}{M + m - m\cos^2(x_3)} \\ \dot{x}_3 &= \dot{\theta} \\ \dot{x}_4 &= \ddot{\theta} = \frac{F\cos(x_3) + (M + m)g\sin(x_3) - mlx_4^2\sin(x_3)\cos(x_3)}{l(M + m - m\cos^2(x_3))} \end{aligned}$$

To analyze the stability near the upright equilibrium, we linearized the system using a Taylor expansion. The upright equilibrium point is

$$x_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \text{ and } F = 0$$

Expanding the function around the equilibrium gives the following

$$\dot{\delta x} = J\delta x + B\delta F$$

where J is the Jacobian matrix of the system and B is the vector of partial derivatives with respect to F both evaluated at the equilibrium. Evaluating these matrices gives

$$J = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & \frac{mg}{M} & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & \frac{(M+m)g}{Ml} & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ \frac{1}{Ml} \end{bmatrix}$$

2.2 Stability analysis

To determine if this upright position is stable, we perform an analysis of the systems eigenvalues. Solving for the eigenvalues gives $\lambda = 0, 0, \sqrt{\frac{(M+m)g}{Ml}}, -\sqrt{\frac{(M+m)g}{Ml}}$. Where M is the mass of the cart, m is the mass of the pendulum bob, g is the gravitational constant, and l is the length of the rod. Additionally, the two zero eigenvalues correspond to the position and velocity of the cart. Since one of the eigenvalues is positive, the system is unstable. This is true because any small change in θ will cause the pendulum to fall. This confirms that the system needs a control mechanism to remain upright, and confirms our model for the system.

3 PID Controller

Now that the system has been linearized around the upright equilibrium point, we can design a PID controller for the pendulum angle. Since we want to keep the pendulum upright, the desired angle is $\theta_d = 0$. Therefore, the error is defined as $e(t) = \theta_d - \theta(t) = -\theta(t)$

Like we mentioned before, the PID controller calculates the force applied to the cart using this equation:

$$F(t) = K_p e(t) + K_i \int_0^t e(\tau) d\tau + K_d \frac{de(t)}{dt}$$

Since $e(t) = -\theta(t)$, this becomes

$$F(t) = -K_p \theta(t) - K_i \int_0^t \theta(\tau) d\tau - K_d \dot{\theta}(t)$$

In the original linearized system, the state vector was $\delta x = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \end{bmatrix}$

However, the PID controller contains an integral term. To include this term in the state-space model, we define a new variable $z(t) = \int_0^t e(\tau) d\tau$. Because $e(t) = -\theta(t)$, we have $\dot{z}(t) = -\theta(t)$. This changes the system from a four-dimensional system into a five-dimensional system with state vector

$$X = \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \\ z \end{bmatrix}$$

Using the linearized model, the uncontrolled system has the form

$$\dot{X} = AX + BF$$

where the fifth row is added to account for the integral error term. The expanded system is

$$\frac{d}{dt} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \\ z \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & \frac{mg}{M} & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & \frac{(M+m)g}{Ml} & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ \dot{x} \\ \theta \\ \dot{\theta} \\ z \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{1}{M} \\ 0 \\ \frac{1}{Ml} \\ 0 \end{bmatrix} F$$

The PID force can now be written in terms of the new state variables

$$F = -K_p\theta - K_d\dot{\theta} + K_i z$$

This controller uses the angle θ , angular velocity $\dot{\theta}$, and accumulated error z to determine the force on the cart. The proportional term responds to the current angle error, the derivative term responds to how quickly the pendulum is falling, and the integral term corrects accumulated error over time (*Inverted Pendulum: System Modeling* 2026).

4 Optimization

After defining the equations and design of the PID controller, the next step is to choose the values for K_p , K_i , and K_d . These constants determine how strongly the controller reacts to the pendulum's angle, accumulated error, and angular velocity respectively.

To compare different PID constants, we must define a cost function. The cost function should be small when the controller keeps the pendulum close to the upright position and large when the pendulum moves far away from equilibrium. A simple cost function is

$$J(K_p, K_i, K_d) = \int_0^T \theta(t)^2 dt$$

This function penalizes the pendulum for moving away from the desired angle $\theta = 0$. Since the main goal of the controller is to keep the pendulum upright, minimizing $\theta(t)^2$ is a natural choice. However, during our first attempt at implementing our gradient descent method for optimization we discovered that this cost function led to inadequate control of the system, specifically it would keep the pendulum mostly upright but at the cost of the cart moving incredibly unrealistically.

Therefore, a more complete cost function is needed. The function we ultimately decided to use is one that includes the cart position and the control force in addition to the angle of the pendulum shown here.

$$J(K_p, K_i, K_d) = \int_0^T (\theta(t)^2 + \alpha x(t)^2 + \beta F(t)^2) dt.$$

In this equation, $\theta(t)^2$ penalizes angular error, $x(t)^2$ penalizes the cart moving too far from its starting position, and $F(t)^2$ penalizes the controller for using too much force. The constants α and β determine how important cart movement and force usage are compared to angle error (Murari, Ahmad, and Kumar 2019).

The optimization problem is therefore

$$\min_{K_p, K_i, K_d} J(K_p, K_i, K_d).$$

The method we chose for finding the best parameters is gradient descent. Starting with an initial guess for the controller constants, gradient descent updates the constants in the direction that lowers the cost function. The update rule is

$$K_p^{new} = K_p - \eta \frac{\partial J}{\partial K_p}$$

$$K_i^{new} = K_i - \eta \frac{\partial J}{\partial K_i}$$

$$K_d^{new} = K_d - \eta \frac{\partial J}{\partial K_d}$$

where η is the learning rate.

Since the cost function depends on the simulated motion of the pendulum, the partial derivatives can be approximated numerically. For example,

$$\frac{\partial J}{\partial K_p} \approx \frac{J(K_p + h, K_i, K_d) - J(K_p, K_i, K_d)}{h},$$

where h is a small step size. The same idea can be used to approximate the derivatives with respect to K_i and K_d .

The optimization process repeatedly simulates the system, calculates the cost, estimates the gradient, and updates the PID constants. The final values of K_p , K_i , and K_d are the values that produce the lowest cost.

When we applied our custom optimization method to our model with the initial parameters

$$\begin{aligned} M &= 1.0kg \\ m &= 0.1kg \\ g &= 9.81m/s^2 \\ l &= 0.5m \\ K_p &= 50 \\ K_i &= 10 \\ K_d &= 20 \end{aligned}$$

The optimized values were found to be

$$\begin{aligned} K_p &= 33.9374 \\ K_i &= 58.9643 \\ K_d &= 7.7545 \end{aligned}$$

These values are very reasonable and are reached with varying initial guess for each parameter indicating that our method is working.

One note about our optimization is that we used the linearized system for computation because it produces cleaner behavior for the gradient that reliably finds a minimum. The effect of this on the accuracy of its findings is minimal because it is calculated very close to the equilibrium where the system was linearized. This accuracy is demonstrated in the next section.

5 Simulation

The initial simulation of the system was done using the linearized equations and with an initial angle of 0.1 rad. We gave our simulator the optimized parameters and allowed the PID controller to attempt to bring the pendulum back to $\theta = 0$. This run is shown below.

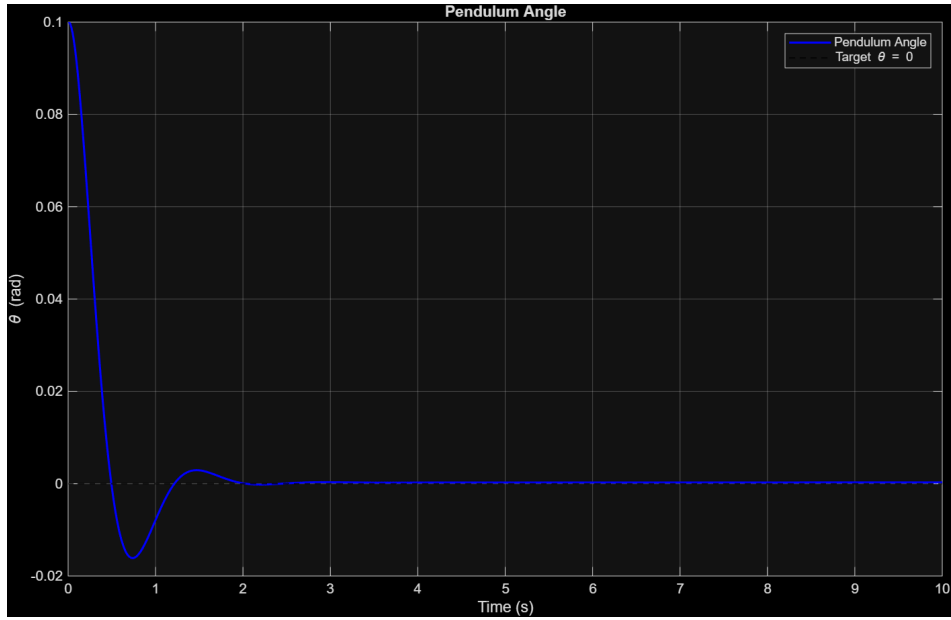


Figure 2: With optimization of K_p , K_i , K_d , the angle is fixed quickly.

Comparing this to the performance of the system with randomly guessed parameters it is clear that the optimization is working as intended.

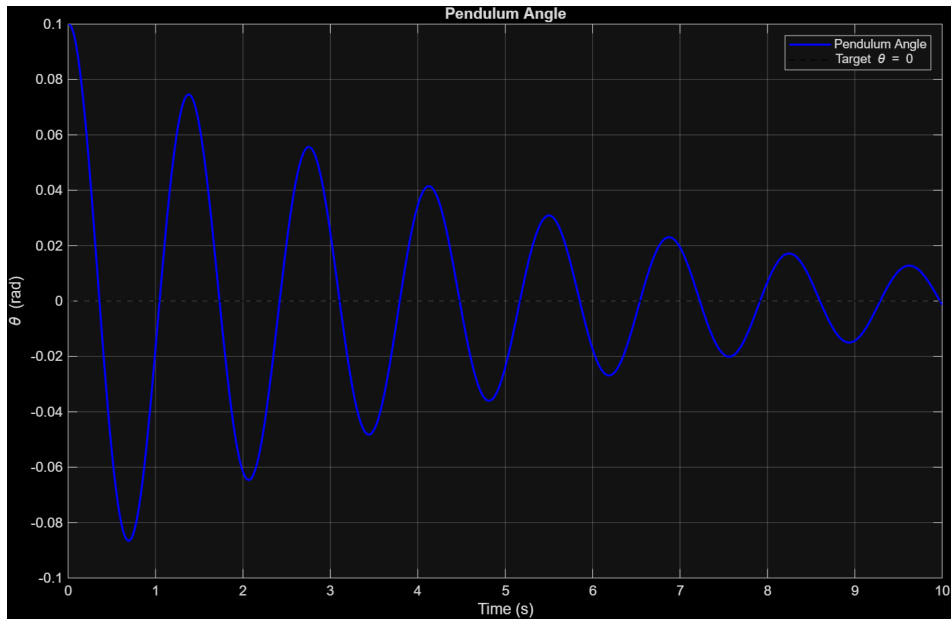


Figure 3: With no optimization of K_p , K_i , K_d there is a lot of oscillation.

To fully test our model it is necessary to introduce a random force that act on the cart and causes the pendulum to become unbalanced, forcing the PID controller to continuously stabilize the pendulum. It is also necessary to use the full, un-linearized, equations of motion for the system to completely model the pendulums movements. Shown below is the pendulum's angel over time with a random force applied to the cart every second modeled by the full equations of motion.

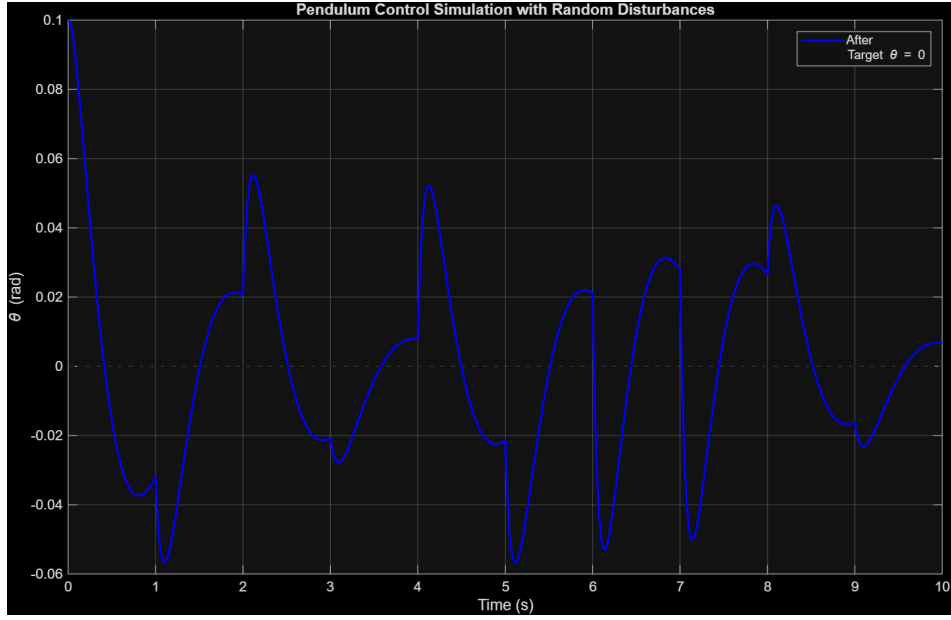


Figure 4: Pendulum angle over time with a random force applied to the cart every second.

Shown below is the force over time graph of the same simulation of the system. It is clear how the force applied to the cart changes in response to the angle of the pendulum.

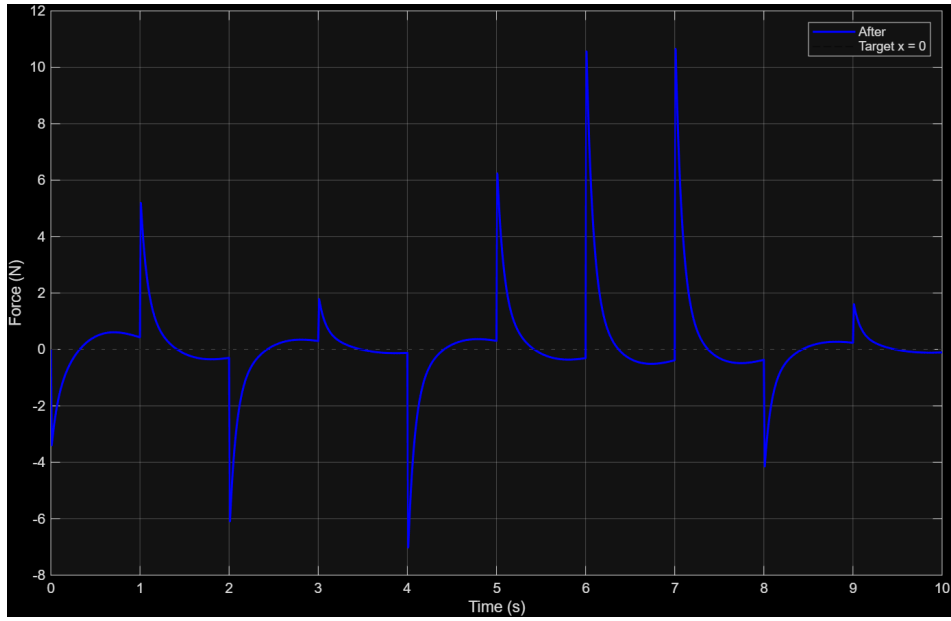


Figure 5: The control force provided to the cart over time.

6 Conclusion

In this paper, we modeled the inverted pendulum on a cart using the Euler-Lagrange equation which yielded a system of 2 nonlinear ODEs which we transformed into a first order system. Linearizing around the upright equilibrium using multivariable Taylor expansion, we were able to confirm the system is unstable without control

To stabilize the system, we designed a PID controller by using the four dimensional state vector of the

system along with a fifth dimension for the integral error term. We found optimal gains by minimizing the cost function using our own implementation of gradient descent with numerical approximation of the partial derivatives. The optimization was performed on the linearized system for simplicity, and the resulting optimized parameters were validated on the full nonlinear system introducing random disturbances, which the controller was able to fix.

It is worth iterating once again that because of the assumptions of our model this is not directly transferable to a real world system, however it can provide a basis for the control of such a system. With a little more work in modeling the system and its constraints, this could provide a better approach to PID tuning than simply random guesses.

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